

A Multi-Arm A/B Testing and Causal Evaluation of Marketing E-Mails with Synthetic Validation and Heterogeneous Treatment Effects

Alexander Yu, Cameron Fong, Nathan Gin
Department of Statistics, University of California, Irvine

December 9, 2025

Contents

1	Introduction	4
2	Background and Motivation	4
3	Data Description	4
4	Exploratory Data Analysis	4
4.1	Treatment Assignment	4
4.2	Outcome Summaries	4
4.3	Covariate Balance	4
4.4	Correlation Structure	4
5	Causal Identification	4
5.1	Unconfoundedness	4
5.2	Overlap	4
5.3	SUTVA	4
5.4	Consistency	4
5.5	Why A/B Tests Identify the ATE	4
6	ATE Results on Hillstrom Dataset	4
6.1	Difference-in-Means	4
6.2	OLS / IPW Estimates	4
6.3	DR-Learner Results	4
6.4	Causal Forest Results	4
7	CATE and Heterogeneous Treatment Effects	4
7.1	CATE by Recency	4
7.2	CATE by Channel	4
7.3	CATE by Purchase History	4
7.4	Causal Forest Feature Importance	4
8	Synthetic Dataset Validation	4
8.1	Synthetic Data Generation	4
8.2	True Treatment Effects	5
8.3	ATE Estimator Comparison	5
8.4	Bias and Variance Evaluation	5
8.5	Interpretation	5
9	Discussion	5
10	Limitations	5
11	Conclusion	5
12	Additional Tables	5
13	Additional Figures	5

- 1 Introduction**
- 2 Background and Motivation**
- 3 Data Description**
- 4 Exploratory Data Analysis**
 - 4.1 Treatment Assignment**
 - 4.2 Outcome Summaries**
 - 4.3 Covariate Balance**
 - 4.4 Correlation Structure**
- 5 Causal Identification**
 - 5.1 Unconfoundedness**
 - 5.2 Overlap**
 - 5.3 SUTVA**
 - 5.4 Consistency**
 - 5.5 Why A/B Tests Identify the ATE**
- 6 ATE Results on Hillstrom Dataset**
 - 6.1 Difference-in-Means**
 - 6.2 OLS / IPW Estimates**
 - 6.3 DR-Learner Results**
 - 6.4 Causal Forest Results**
- 7 CATE and Heterogeneous Treatment Effects**
 - 7.1 CATE by Recency**
 - 7.2 CATE by Channel**
 - 7.3 CATE by Purchase History**
 - 7.4 Causal Forest Feature Importance**
- 8 Synthetic Dataset Validation**
 - 8.1 Synthetic Data Generation**

To complement our analysis of the real e-commerce A/B test, we generated a synthetic dataset in which the full data-generating process is explicitly controlled. In the real dataset, treatment assignment and potential

outcomes are not fully observable, so we cannot assess whether our estimated ATEs and CATEs are accurate or biased. Our synthetic dataset allows us to specify exactly how treatment depends on customer covariates, including the confounders such as recency, purchase history, gender-aligned preferences, and newbie status. It also allows us to insert known causal effects into the outcome model, ensuring that the true treatment effects are known by construction. With this structure, we can evaluate how well different causal inference methods recover the correct treatment effects by directly computing their estimation bias and variance. This provides a controlled validation scenario that mirrors the patterns observed in our real data while giving us the ability to assess method performance, which is not possible when analyzing the actual experiment alone.

8.2 True Treatment Effects

8.3 ATE Estimator Comparison

8.4 Bias and Variance Evaluation

8.5 Interpretation

9 Discussion

10 Limitations

11 Conclusion

12 Additional Tables

13 Additional Figures